Nonnegative and positive factorizations

# Connectedness of minimal positive semidefinite factorization orbits

**Difficulty:** challenging **Importance:** interesting to the community

**Status:** open; checked 2026-09-08
**Area:** geometry of constrained matrix factorizations

## Context and notation

Write $\mathbb{S}^k_+$ for the cone of real symmetric positive semidefinite $k \times k$ matrices. For an entrywise nonnegative matrix $M \in \mathbb{R}^{p\times q}_+$, its **real positive semidefinite rank** is

$$\mathrm{rank}_{\mathrm{psd}}(M) = \min\{k \geq 1 : \exists A_1, \dots, A_p, B_1, \dots, B_q \in \mathbb{S}^k_+, \quad M_{ij} = \mathrm{tr}(A_i B_j) \text{ for all } i, j\}.$$

This definition concerns a family of matrix factors indexed by the rows and columns of $M$.

## Problem statement

Let $k \geq 3$ and $p, q \geq 1$ be integers. Suppose $M \in \mathbb{R}^{p\times q}_+$ satisfies

$$\mathrm{rank}(M) = \frac{k(k+1)}{2}, \qquad \mathrm{rank}_{\mathrm{psd}}(M) = k.$$

Let $\mathcal{F}_k(M)$ be the set of all tuples $(A_1, \dots, A_p, B_1, \dots, B_q)$ in $(\mathbb{S}^k_+)^{p+q}$ satisfying $\mathrm{tr}(A_i B_j) = M_{ij}$ for every $i, j$. Give it the Euclidean subspace topology. Identify tuples under the changes of basis

$$A_i \longmapsto S^{\mathsf{T}} A_i S, \qquad B_j \longmapsto S^{-1} B_j S^{-\mathsf{T}}, \qquad S \in GL(k, \mathbb{R}).$$

Give the resulting orbit space the quotient topology.

### Question

Is $\mathcal{F}_k(M)/GL(k, \mathbb{R})$ connected for every $M$ satisfying these hypotheses?

The ordinary-rank hypothesis is part of the problem. The case $k = 2$ is known to be connected. The question asks whether optimal factors can belong to separated families after changes of basis are identified.

## References

Fawzi, Gouveia, Parrilo, Robinson, and Thomas, *Positive semidefinite rank*, §9.2, Problem 9.4. Richard Z. Robinson, *The Positive Semidefinite Rank of Matrices and Polytopes*, University of Washington dissertation (2015), Chapter 7, especially Proposition 7.0.8.

## Status evidence

Searches for "psd factorizations connected", "positive semidefinite factorization space connected", and "psd factorization orbits connected 2026" found the original question and the known $k = 2$ result, but no resolution under the displayed rank conditions. Universality or disconnectedness results for nonnegative vector factorizations do not by themselves answer this question. No recent explicit open-status confirmation was located.